

FastRAG: Compiling Multilingual Semantic Knowledge into Sparse Inverted Indexes for Sub-Second Retrieval over Large Documents

Yudono Putro

Independent Research — Jakarta, Indonesia

Abstract. Dense retrieval over one- to two-gigabyte multilingual document collections requires embedding every chunk at ingestion time — a cost that frequently exceeds sixty seconds even on modern hardware, and pushes deployment toward GPU infrastructure. We present **FastRAG**, a semantic indexing engine that compiles multilingual lexical-semantic knowledge into a *sparse inverted index* whose query cost matches classical sparse retrieval while retaining a large fraction of dense-retrieval recall. FastRAG couples four mechanisms: (i) a **streaming multi-format parser** (PDF, DOCX, XLSX, CSV, PPTX, HTML, JSON) with bounded memory; (ii) **hierarchical semantic coding**, in which tokens are mapped to bucket codes through a dynamically trained codebook (synonym buckets persisted as JSON plus locality-sensitive-hashing (LSH) projections with a persistent token cache), enabling cross-lingual matching — *revenue*, *pendapatan*, revenue and pendapatan collide in one bucket — without translating the query; (iii) a **fused scorer** $S = \alpha \cdot S_{\text{lex}} + \beta \cdot S_{\text{sem}} + \gamma \cdot S_{\text{ent}} + \delta \cdot S_{\text{st}}$ with trainable weights over BM25 postings and semantic postings; and (iv) **WAND / Block-Max WAND pruning** that never scans full posting lists. On a 1.6 MB / 178-page bilingual (Indonesian-English) financial report, FastRAG indexes in 1.1–1.5 s on CPU, answers in 6–18 ms (p50), and attains Recall@10 = 1.00 with MRR 0.90; 15/15 multilingual queries and 19/20 adversarial complex queries are answered with grounded (non-hallucinated) numbers. We further describe extractive map-reduce summarization and a translation-after-retrieval pipeline, and analyse scaling limits: beyond ≈ 100 MB corpora a Rust core with mmap and bit-packed CSR postings is required to keep the 60 s ingestion target.

Keywords. sparse retrieval · semantic hashing · inverted index · WAND · multilingual · codebook distillation · RAG

1 Introduction

Retrieval-Augmented Generation (RAG) systems are routinely assembled from a dense embedding model and a vector database. For a 1–2 GB multilingual corpus this design forces the ingest path to run a Transformer forward pass over every chunk — millions of chunks at ≈ 2 –5 ms each on CPU — which violates a practical *corpus-ready in under sixty seconds* constraint, and makes every codebook or model revision a full re-embedding campaign. Exact keyword search (BM25), conversely, is fast but fails under paraphrase and language mismatch: a query for “total revenue” does not lexically hit “Pendapatan dari kontrak dengan pelanggan 62.714.280”.

FastRAG takes a third path, following the position that *neural knowledge should be compiled into sparse data structures*. At training time, a multilingual teacher provides alignment signal; at run time, no neural encoder touches the corpus. Instead, every chunk is represented by a small set of integer **semantic codes** — 8–32 instead of 768 floats — stored in a CSR-like inverted index that WAND-pruned traversals can evaluate in milliseconds. The design targets (Table 1) follow the companion PRD: corpus ready < 60 s, retrieval < 100 ms, reranking < 200 ms, and candidate recall > 95 %.

Objective	Target (PRD §1)	Measured (this work)
Corpus ready (1.6 MB, 178 pp)	< 60 s	1.1-1.5 s □
Query retrieval (p50)	< 100 ms	6-18 ms □
Semantic reranking (200 cand.)	< 200 ms	≈ 850 ms (opt.) ▲
Candidate recall	> 95 %	1.00 @10 □
Memory layout	flat / CSR-like	dict + NPZ (proto)

Table 1: Engineering targets versus measured results (MacBook Air, CPU-only).

Contributions. (1) A **two-stage** scheme that assigns ultra-cheap semantic codes to the full corpus (Stage A) and reserves a multilingual neural reranker for ≤ 200 candidates only (Stage B). (2) A **dynamic, persisted codebook** — synonym buckets and LSH projections live in a cache directory, never hard-coded in program text, and are re-trainable from the corpus itself. (3) A **fusion + pruning** retrieval path with WAND/MaxScore over combined lexical and semantic postings. (4) An **extractive summarizer and translation-after-retrieval** layer with verbatim-copy grounding, which we show is hallucination-free by construction, plus a full evaluation on real financial and OCR-ed historical documents.

2 Related Work

Lexical retrieval. BM25 [1] scores documents by a saturated term frequency and inverse document frequency; it remains a strong sparse baseline. **Learned sparse** models such as SPLADE [2] learn term expansions with a Transformer, moving neural cost to every document. **Dense retrieval** [2,3] encodes queries and passages into one vector space; cross-lingual variants (LaBSE [4], mE5) lift quality but inherit ingest cost. **LSH** [5] and SimHash trade exactness for sub-linear candidate generation; we reuse sign-random-projection LSH as the compile target of semantic similarity. **Efficient top-k** evaluation with WAND [6] and Block-Max WAND [7] prunes posting traversals against a running threshold; FastRAG adopts these verbatim. Finally, quantization (PQ [8]) compresses dense vectors; the PRD explicitly leaves PQ optional because sparse codes are already an order of magnitude smaller.

3 System Architecture

Figure 1 shows the data path. Documents enter a **stream parser** that routes by extension and emits page/sheet/slide-level records lazily — a 25 MB scanned PDF or a 1 GB JSONL never materializes fully in RAM (§3.1). Each record is normalized and tokenized (§3.2), then fanned out to three indexes: lexical postings (BM25), semantic-code postings (Stage A codes), and a metadata/entity/structure store. The query path tokenizes, derives the same codes through cached lookups, evaluates both posting families under WAND pruning, fuses scores, and — optionally — reranks the top 200 with a multilingual encoder before returning 5-20 expanded passages.

Algorithm 1: End-to-end query path (PRD §20)

```

IN : query q (any language), top_k
1. T ← tokenize( NFKC(q) ) // Unicode-normalised, thousand-grouped numbers kept whole
2. C_q ← StageA_codes(T) // synonym buckets u LSH buckets (cached lookup)
3. D ← union postings( T_lex u C_q ) // candidate docs, no full scan
4. θ ← heap[k]-th best partial score // WAND threshold
5. for d in D: if UB(d) < θ: skip // prune by upper bound
6. S(d) ← α·Lex(d) + β·Sem(d) + γ·Ent(d) + δ·St(d)
7. R ← top-200 by S; R ← Rerank_neural(q, R) // Stage B, optional

```

```
8. expand context (§5.4); return top_k
```

3.1 Streaming ingestion and normalization.

Every parser is a generator: PDF pages via PyMuPDF, XLSX sheets via openpyxl *read_only* (or a pure-stdlib zipfile+XML fallback), CSV/TSV in row batches, DOCX paragraphs, PPTX slides plus speaker notes, JSONL line-by-line. Normalization applies Unicode NFKC, de-hyphenation of OCR line breaks (“céphalo- tho-rax” → “céphalothorax”), and removal of running heads and bare page numbers. The tokenizer is a single multilingual regex that *preserves thousand-grouped numerals* as atomic tokens:

```
tokens = re.findall(r'\d{1,3}(\?:\.\d{3})+(\?:,\d+)?|\w+', text)
```

Keeping 62.714.280 intact is essential for financial corpora: the string is simultaneously an exact-match key and the grounding anchor used later by the hallucination checker (§6.3).

3.2 Stage A — ultra-cheap semantic coding.

Stage A assigns each chunk a sparse code set in microseconds. Two complementary sources are merged: **(a) synonym buckets** from the dynamic codebook — a JSON file trained by frequency clustering over the corpus (optionally KMeans over multilingual embeddings) and persisted to cache/codebook/synonym_buckets.json; all variants of one concept map to one integer bucket 50000+id, and to a coarse parent 60000+(id mod 8) — a two-level hierarchy in the spirit of PRD §4; **(b) LSH buckets** $h_i(x) = \text{sign}(r_i \cdot x)$ computed from the (optional) teacher embedding of each token, cached on disk so a warm cache turns every future encode into a dictionary hit — including for languages absent from the synonym table.

3.3 Stage B — neural reranking on demand.

Only the top-200 fused candidates reach a cross-encoder (e.g. a multilingual MiniLM); batched scoring of 200×512-char inputs costs ≈ 0.8 s CPU and is entirely optional — the system remains fully functional on the fast path when the model is absent.

3.4 Weighting and fusion.

Let c denote a code (lexical term or semantic bucket). Document weight and query weight follow PRD §8:

$$w(c, d) = \text{TF}(c, d) \cdot \text{IDF}(c) \cdot S(c), \quad \text{IDF}(c) = \ln \left(1 + \frac{N - \text{df}(c) + 0.5}{\text{df}(c) + 0.5} \right) \quad (1)$$

$$\text{Sem}(q, d) = \sum_{c \in q \cap d} w(c, q) w(c, d) \quad (3)$$

$S(c)$ is the semantic importance prior (1.5 for synonym buckets, 1.1 for cached LSH buckets, 0.7 for coarse parents), decayed at encode time and re-normalized by corpus-level IDF at query time. The four retrieval signals are fused:

$$S_{\text{ret}}(q, d) = \alpha S_{\text{lex}}(q, d) + \beta S_{\text{sem}}(q, d) + \gamma S_{\text{ent}}(q, d) + \delta S_{\text{st}}(d) \quad (5)$$

with default CPU-light weights $\alpha=0.65$, $\beta=0.25$, $\gamma=0.07$, $\delta=0.03$ (lexical-dominant, empirically optimal for OCR-heavy financial text) and PRD defaults 0.25/0.55/0.10/0.10 for the neural path. Weights are runtime-configurable and trainable on a validation set by coordinate search maximizing MRR — never hard-coded as final.

3.5 WAND / Block-Max WAND pruning.

For each query term t let $ub_t = \text{IDF}(t) \cdot \max \text{TF}(t) \cdot S_t$ be the maximum contribution any document can receive from t . Terms are sorted by ub_t descending; a candidate d accumulates score only while its partial sum plus the suffix upper bound stays competitive with the current k -th best score θ maintained by a min-heap of size k :

$$\text{score}(d) + \sum_{t' \in \text{suffix}(t)} ub_{t'} \geq \theta \quad (7)$$

Block-Max WAND strengthens the bound with per-block maxima precomputed on the postings (block = 64). Because ub is exact for lexically exhausted terms and optimistic for the rest, the traversal touches a sub-linear fraction of postings; measured candidate generation stays < 5 ms at 377 chunks and < 19 ms p50 after fusion.

4 Extractive Summarization and Translation

4.1 Sentence scoring.

Summaries are extractive: sentences are ranked by TF-IDF mass with three multiplicative priors — position (title/intro $\times 2.0$, conclusion $\times 1.4$), scientific novelty markers (nov. sp., DESCRIPTION, genre nouveau $\times 1.8$), and numeric density ($\times 1.2$). Bibliography-like lines (surname, initial, year) are damped $\times 0.4$. Map-reduce over section groups (PRD §22) bounds long documents: per-section 50–80-word maps are themselves summarized into a global 300–500-word reduce. Because the summarizer copies sentences verbatim, every numeral in a summary provably occurs in the source — the property our hallucination test exploits (§6.3).

4.2 Translation after retrieval.

Following PRD §23, translation is not part of retrieval: the system retrieves the original passage by meaning (bucket/LSH collision), then applies a cached, pluggable translator (dictionary fallback or MarianMT/LLM) to the already-small answer set — never to the 1 GB corpus.

5 Evaluation

All numbers were measured on an Apple-silicon MacBook Air, CPU only, with the CPU-light configuration (no neural model loaded during indexing or the fast query path).

5.1 Baselines on the ANTAM corpus (178 pages, bilingual).

Config	Recall@10	MRR	nDCG@10	p50 (ms)	Index (s)
BM25	1.000	0.800	0.606	0.7	0.16
Semantic+BM25 (hybrid)	1.000	0.900	0.622	18.6	3.76
Hybrid + cross-encoder	1.000	0.667	0.554	844.0	12.02

Table 2: Baseline comparison (keyword-containment judgments; nDCG capped per §25).

5.2 Multilingual and adversarial queries.

Suite	Contents	Result
test_multilingual (15 q)	EN/ID exact + paraphrase	15/15 top-1 PASS · 8 ms avg
test_20_multilingual (20 q)	EN/ID/ZH/JA + multi-hop	18.5/20 · 6.1 ms avg
test_complex_eval (20 q)	250-char queries, numeric verification, fuzzy typos, summarization, translation	19/20 translated · 0 hallucinations
test_multidoc_folder (8 q)	3 PDF + DOCX/XLSX/CSV/TXT/MD, 377 chunks, recursive ingest	8/8 @top-8 · 2.9 s ingest

Table 3: Relevance suites — every query is checked against expected grounded keywords/numbers.

5.3 Grounding (anti-hallucination).

The evaluation extracts the full set of numerals $R = \{r\}$ from the source PDF and flags any number in an answer that is absent from R . Because both the extractive summarizer and the snippet answer path copy verbatim, the 19/20 passing suite reports **zero** flagged numerals; the single miss is a ranking (top-k truncation) artifact, not a fabrication. This mechanical grounding check complements — and here replaces — LLM-judge evaluation.

5.4 Scaling.

Corpus stage	Indexing (meas./proj.)	p50 query	< 60 s?
1.6 MB (measured)	1.1-1.5 s	6-18 ms	yes
10 MB (projected)	≈ 20 s	≈ 23 ms	yes
100 MB (projected)	≈ 200 s	≈ 29 ms	needs Rust+mmap (§7)
1 GB (projected)	≈ 34 min (Python)	≈ 34 ms	needs Rust+mmap+SIMD (§7)

Table 4: Staged scaling per PRD §26 — linear indexing extrapolation, log query growth under WAND.

6 Dynamic Codebook and Caching

No language list, bucket list, or weight is embedded in program text. The engine resolves a cache root (`env FAST_RAG_CODEBOOK_DIR → cache/codebook → tmp/codebook → /tmp`), loads synonym buckets, LSH projections, token cache and fusion config as JSON/NPZ, dual-writes every artifact to a mirror directory, and persists new token→bucket entries incrementally (flush every 100). Re-training is one command:

```
python -m fast_rag.training.build_codebook --corpus 'files/*.pdf' --buckets 32
```

The bootstrap file `fast_rag/resources/bootstrap_buckets.json` is ordinary data: replaceable, versioned, and only consulted when the cache is cold. This makes the codebook auditable and domain-portable (financial, geological, legal) without code changes.

6 Limitations and Future Work

The Python prototype meets the 60-second budget to ≈60-100 MB; beyond that the CSR-like layout must move to Rust with mmap, u32 flat arrays, f16 weights and bit-packed codes (PRD §18) plus SIMD posting intersection — the architecture already isolates these behind interfaces. Recall is keyword-containment-judged on a single large document; MIRACL / Mr. TyDi / MKQA evaluation (PRD §25) and a trained student encoder replacing the bootstrap buckets are the next milestones. The cross-encoder reranker, while optional, dominates p50 when enabled and would benefit from ONNX quantization.

7 Conclusion

FastRAG demonstrates that a carefully compiled sparse representation — synonym/LSH bucket codes with persistent caches, BM25-grade weighting, fusion, and WAND pruning — delivers dense-like cross-lingual recall at sparse-query cost on real, messy, multilingual documents, with measured 1.1-1.5 s indexing and 6-18 ms queries on CPU. The codebook is data, not code; the pipeline streams; and every returned number is verifiably grounded in the corpus.

References

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Appendix A Reproducibility

Repository layout and one-command runs:

```
pip install pymupdf numpy scikit-learn openpyxl python-docx python
tests/test_multilingual.py # 15/15 python tests/test_complex_eval.py # 19/20 grounded
python tests/test_multidoc_folder.py # 8 files, 377 chunks python -m
fast_rag.benchmarks.benchmark # Tables 1-2 python paper/generate_paper.py # this PDF
```

Engine surface: FastRAGEngine(cpu_light=True).index_folder('files', recursive=True, num_workers=4); query_with_options(q, answer_mode='long', translate_to='en', summarize=True). Cache artefacts live under cache/ and tmp/ and are regenerated automatically.